

3. Distributed computing:

- Multiple computers worked together to solve large computational tasks.
- Workloads were distributed systems, improving performance and reliability.
- Examples include scientific research and large scale simulations.

4. Grid computing:

- Grid computing connected computers from different locations to share processing power.
- Resources were used collectively for complex tasks.
- This became the foundation of modern cloud computing.

5. Cloud computing

- Computing resources such as servers, databases and software are delivered over the internet.
- Users can access services anytime, anywhere, on demand.
- Cloud providers manage the infrastructure while users pay only for what they use.
- Examples include Google cloud, Microsoft Azure and Amazon web services (AWS).

Q2 Interpret the working of hypervisor in virtualization
Different b/w Type 1 and Type 2 hypervisors with examples

Hypervisor in virtualization:

A hypervisor (is software that creates and manages virtual machine. It allows multiple operating systems to run simultaneously on a single physical computer by sharing hardware resources such as CPU, memory, storage and network.

Working of a hypervisor

- The physical computer starts.
- The hypervisor is loaded
- It allocates hardware resources (CPU, RAM, storage, network) to each virtual machine.
- Each VM runs its own operating system internally.
- Multiple VMs can run simultaneously without affecting each other.

Advantages:

- Better hardware utilization.
- Reduced infrastructure cost.
- Isolation between virtual machines.
- Easy backup migration and recovery.

Type 1 Hypervisor

- Installed directly on the physical server.
- No host operating system is required
- provides better speed, security and reliability
- Commonly used in cloud computing and enterprise environments.

Examples:

- VM ware ESxi
- Microsoft Hyper-V
- Xen Hypervisor

Type 2 Hypervisor

- Installed as an application on an existing operating system.
- Depends on the host OS to access hardware
- Easy to install and suitable for development testing and education.

Examples:

Oracle Virtual Box

VM ware Workstation

parallels Desktop.

3 Describe the role of APIs in cloud service integration. Explain how APIs support interoperability b/w services.

* Role of APIs in cloud service integration

An Application Programming Interface (API) is a set of rules and protocols that allows different software applications to communicate with each other. In cloud computing, APIs enable applications, platforms and cloud services to exchange data and perform operations seamlessly.

Roles of APIs in cloud integration.

- Connect different cloud application and services.
- Enable secure data exchange b/w systems.
- Automate cloud resource provisioning and management.
- Allow developers to integrate third-party services into applications.

How APIs support Interoperability b/w services.

Interoperability means the ability of different systems, applications or cloud platforms to work together and exchange information effectively.

1. standard communication.

- APIs use standard protocols such as HTTP and HTTPS and data formats like JSON and XML.
- This allows applications built with different programming languages to communicate easily.

2. Data sharing:

- APIs connect services from different cloud providers.
- Example: An e-commerce website stores product images in cloud storage and processes payments using third party payment API.

3. Automation:

- APIs automate tasks such as creating virtual machines, managing storages, monitoring resources and deploying applications.

4. Security:

APIs use authentication methods such as API keys, OAuth and access tokens to ensure only authorized users and applications can access cloud resources.

Explain the concept of Resource Pooling in cloud computing. Illustrate how it Benefits Multiple Users.

A Resource Pooling in cloud computing:

cloud service providers use a multi-tenant model where the same physical infrastructure is shared among multiple customers while keeping each customer's data secure and isolated.

How Resource Pooling works.

1. cloud providers maintain a large pool of physical resources in data centers.
2. These resources are virtualized using virtualization technology.
3. When a user requests a service the cloud allocates the required resources from the shared pool.
4. After the user finishes the resources they are returned to the pool and made available to other users.
5. Resources are automatically assigned and reallocated according to demand.

Benefits of Resource pooling for Multiple Users:

1. Efficient Resource Utilization:

- Resources are shared among many users, reducing idle hardware and maximizing utilization.

2. Cost Reduction:

- Users pay only for the resources they consume, reducing infrastructure and maintenance costs.

3. Scalability:

- Resources can be increased or decreased instantly based on user demand.

4. High Availability:

- Shared infrastructure ensures continuous service even if some hardware components fail.

5. Flexibility

- Users can compute resources anytime without purchasing physical hardware.

6. Easy Management:

- The cloud provider manages the hardware updates, and maintenance allowing users to focus their applications.

Interpret the Concept of Rapid Elasticity in cloud computing. Explain how it supports dynamic workloads.

Rapid Elasticity in cloud computing:

Rapid elasticity is one of the key characteristics of cloud computing. It refers to the ability of cloud systems to quickly increase or decrease computing resources based on user demand.

How Rapid Elasticity works:

1. A user deploys an application on the cloud.
2. The cloud continuously monitors the application's resource usage.
3. When demand or traffic increases additional resources are automatically allocated.
4. When demand decreases unused resources are released back to the resource pool.
5. Users are charged only for the resources they actually use.

How Rapid Elasticity supports Dynamic workloads:

i. Automatic scaling:

Resources automatically increase during high demand and decrease during low demand without

manual intervention.

2. Handles Traffic spikes:

Applications continue to perform efficiently during sudden increases in users or requests.

3. Cost Efficiency

• Users continue to perform efficiently during sudden increase in infrastructure costs.

4. Improved performance:

Additional resources prevent slow response times and maintain smooth application performance.

5. Business continuity:

Applications remain available even during peak ensuring uninterrupted services

6. Flexibility:

organizations can easily adapt to changing business requirements without purchasing new hardware